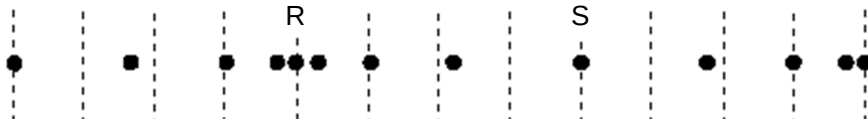


- 16** The figure shows the positions of particles of a medium at a particular instant when a longitudinal wave, travelling from left to right, passes through the medium. Before the wave arrived, the particles were all spaced equally apart on the vertical dotted lines shown. At the instant shown, particles R and S are passing through their original undisturbed positions.



Which of the following describes the direction of the velocities of particles R and S at the instant shown?

	velocity direction for R	velocity direction for S
A	right	left
B	right	right
C	left	left
D	left	right

- 17** A person, along with a medium, is moving at 25 m/s from an observer of an object